

DSA210 Project Report

Conflict, Oil Prices, and Economic Performance in the Middle East

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GitHub Repository: <https://github.com/HussamAlabrash/DSA210PROJECT>

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1. Introduction

This project analyzes the relationship between conflict, oil prices, and economic performance in Middle Eastern countries. The Middle East is an important region for studying economic volatility because many countries are affected by oil dependency, regional conflict, political instability, and external economic shocks. The main goal of this project is to investigate whether conflict-related indicators and oil price movements are associated with macroeconomic outcomes such as GDP growth, inflation, and foreign direct investment.

The project follows the full data science pipeline. First, two datasets are collected and merged. Then, exploratory data analysis is performed using multiple visualizations. After that, hypothesis testing is applied to examine relationships between key variables. Finally, supervised machine learning models are trained and evaluated to predict annual GDP growth.

2. Motivation

The motivation behind this project is to better understand how instability and oil price fluctuations affect economic performance in the Middle East. Many countries in the region rely heavily on oil revenues, while others are strongly affected by conflict and political uncertainty. These factors can influence GDP growth, inflation, foreign investment, unemployment, and trade.

By combining economic data with conflict event data, this project aims to identify patterns that may not be visible when looking at only one dataset. The project is also motivated by the broader question of whether data science methods can help explain and predict economic outcomes in politically unstable regions.

3. Data Overview

The analysis uses two publicly available Kaggle datasets.

1. **Middle East Economic Data 1990-2024 with Oil**

This dataset includes country-year economic indicators such as GDP growth, GDP per capita, GDP current USD, inflation, imports, exports, unemployment, foreign direct investment, life expectancy, and Brent oil price.

2. **MiddleEast 2015-2024 Nov01 Conflict Dataset**

This dataset includes conflict-related observations in Middle Eastern countries. The dataset contains event-level conflict data, including conflict event identifiers, country, year, and fatalities.

The economic dataset was filtered to the overlapping period between 2015 and 2024. The conflict dataset was aggregated by country and year to create two main conflict variables: total conflict events and total fatalities. The two datasets were then merged using country and year.

4. Methodology

The project methodology consisted of the following steps:

- Load the economic and conflict datasets into Python.
 - Aggregate the conflict dataset by country and year.
 - Filter the economic dataset to the years 2015-2024.
 - Clean country names to avoid merge mismatches.
 - Merge the datasets using country and year.
 - Fill missing conflict event and fatality values with zero after merging.
 - Conduct exploratory data analysis using several types of graphs.
 - Perform hypothesis testing using Pearson correlation.
 - Apply supervised machine learning regression models to predict annual GDP growth.
 - Compare model performance using MAE, RMSE, and R2 score.
 - Interpret the model results and feature importance.
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5. Data Collection and Preparation

The economic dataset and conflict dataset were collected from Kaggle and uploaded into Google Colab. Since the two datasets had different structures, preparation was necessary before analysis.

The conflict dataset was originally at the event level, meaning that each row represented a conflict event. To merge it with the economic data, the conflict data was grouped by country and year. The number of conflict events was counted using the event identifier, and total fatalities were calculated by summing the fatalities column.

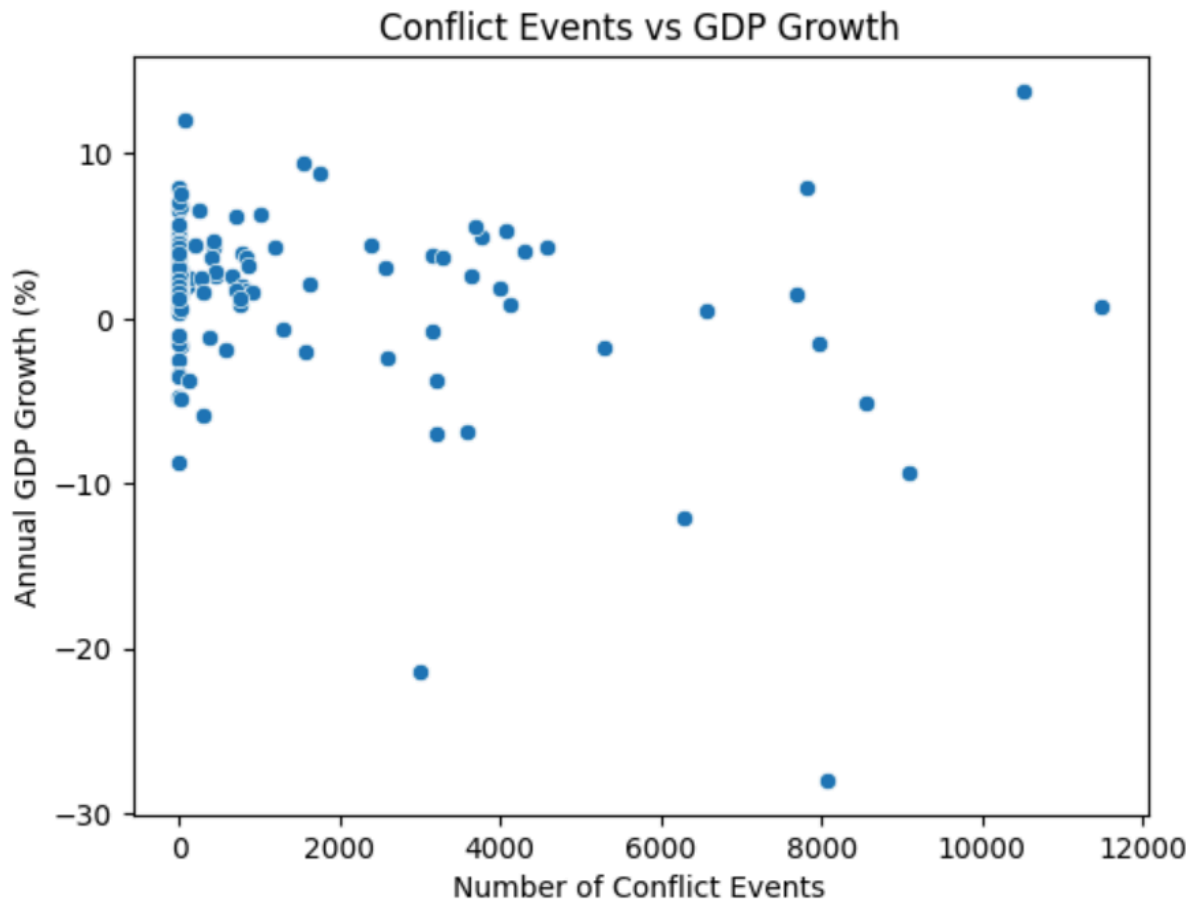
Country names were cleaned by removing extra spaces and converting them to lowercase. This was important because small differences in spelling or capitalization could prevent rows from matching correctly during the merge. After merging, missing conflict values were filled with zero because a missing conflict entry after the merge indicated no recorded conflict event for that country-year in the conflict dataset.

The final merged dataset contained economic indicators, oil price, conflict events, and fatalities for Middle Eastern countries between 2015 and 2024.

6. Exploratory Data Analysis

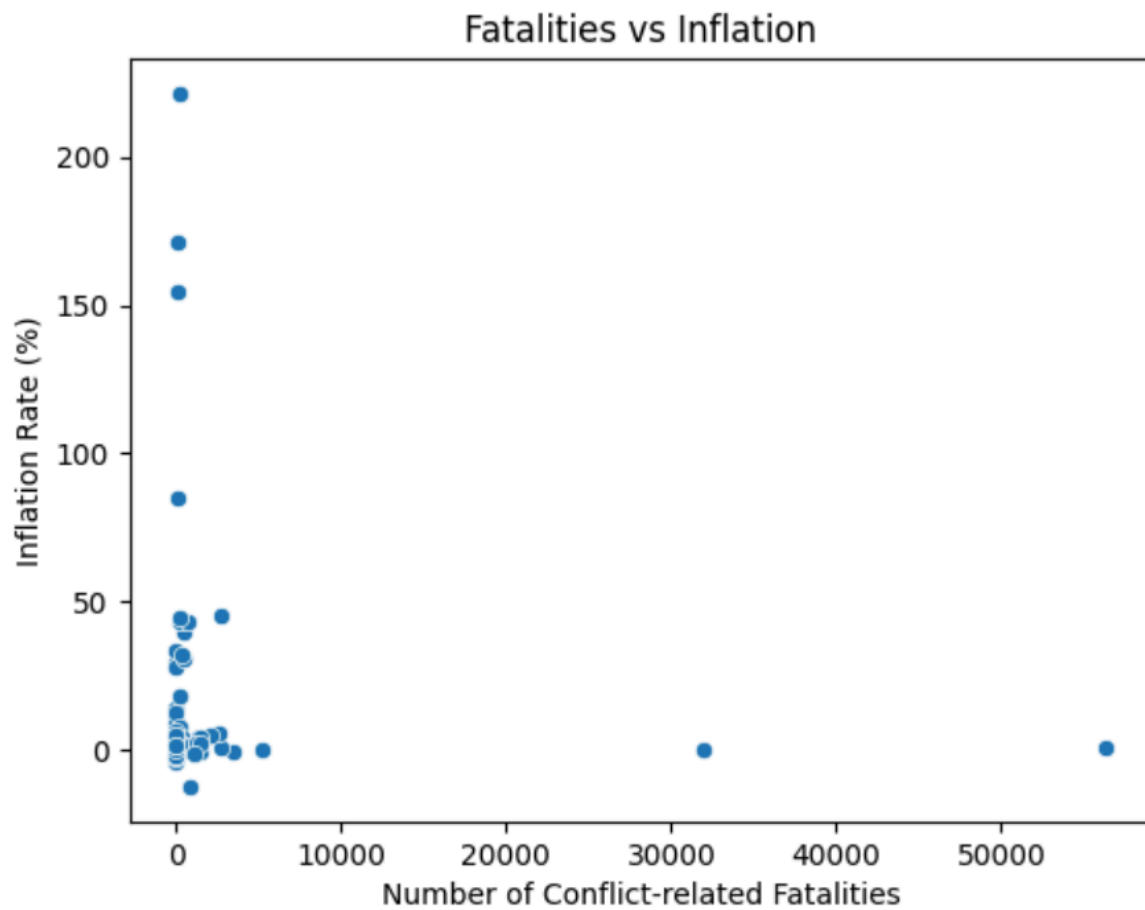
Exploratory data analysis was conducted to understand the structure of the merged dataset and identify patterns before hypothesis testing and machine learning. Multiple plot types were used, including scatter plots, histograms, boxplots, line plots, bar charts, and a correlation heatmap.

6.1 Conflict Events vs GDP Growth



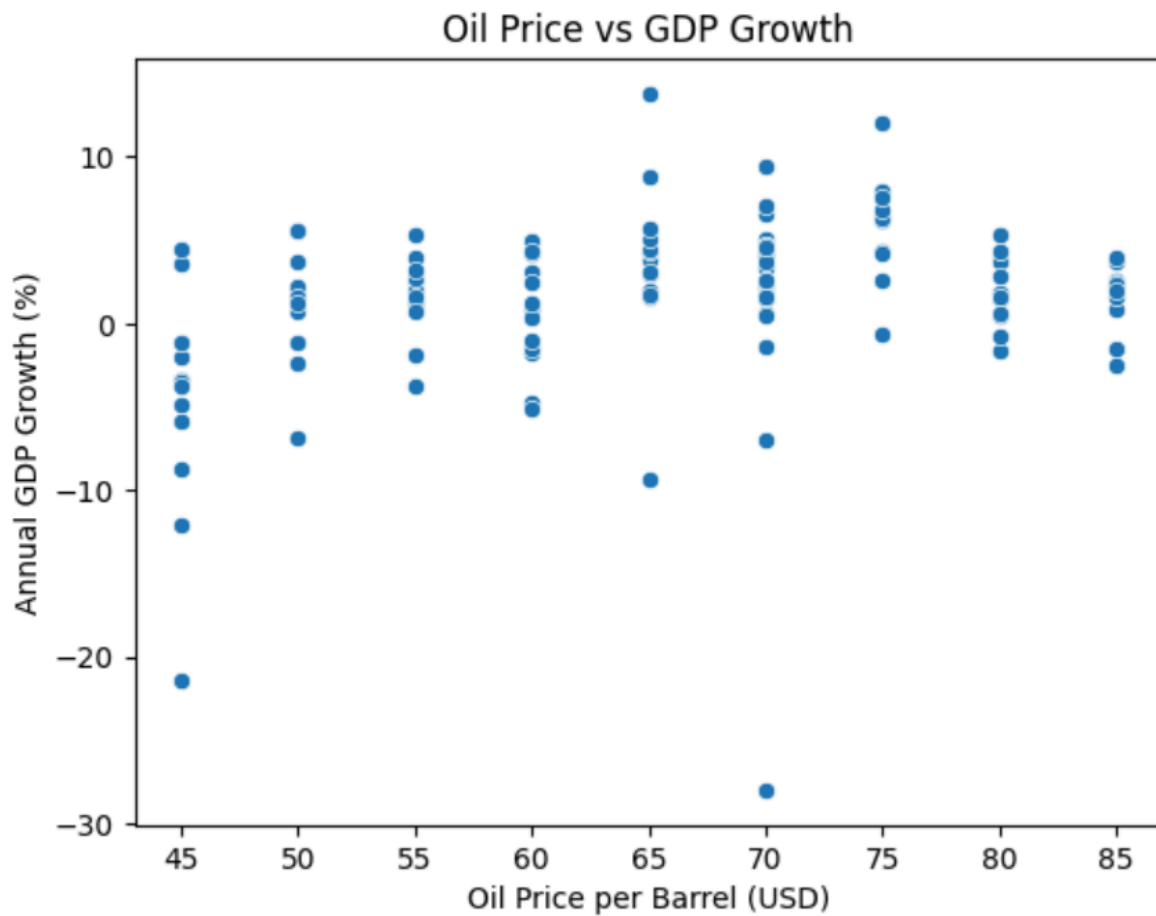
This scatter plot examines the relationship between the number of conflict events and annual GDP growth. The points are widely dispersed, meaning the relationship is not perfectly linear. However, higher conflict event values are often associated with more volatile and sometimes lower GDP growth. This suggests that conflict may contribute to economic instability.

6.2 Fatalities vs Inflation



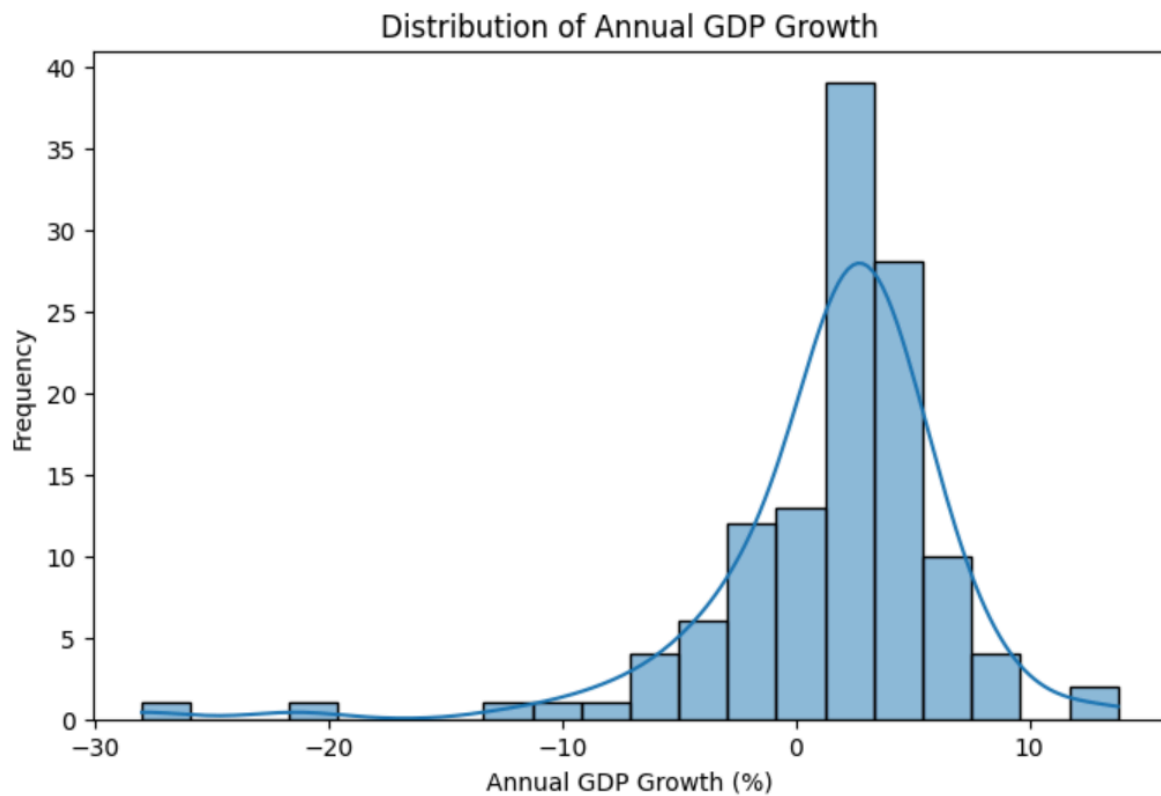
This plot shows the relationship between conflict-related fatalities and inflation. The observations do not show a strong clear trend. This suggests that fatalities alone may not directly explain inflation changes in the dataset. Inflation is likely affected by other factors such as monetary policy, exchange rates, supply shocks, and import costs.

6.3 Oil Price vs GDP Growth



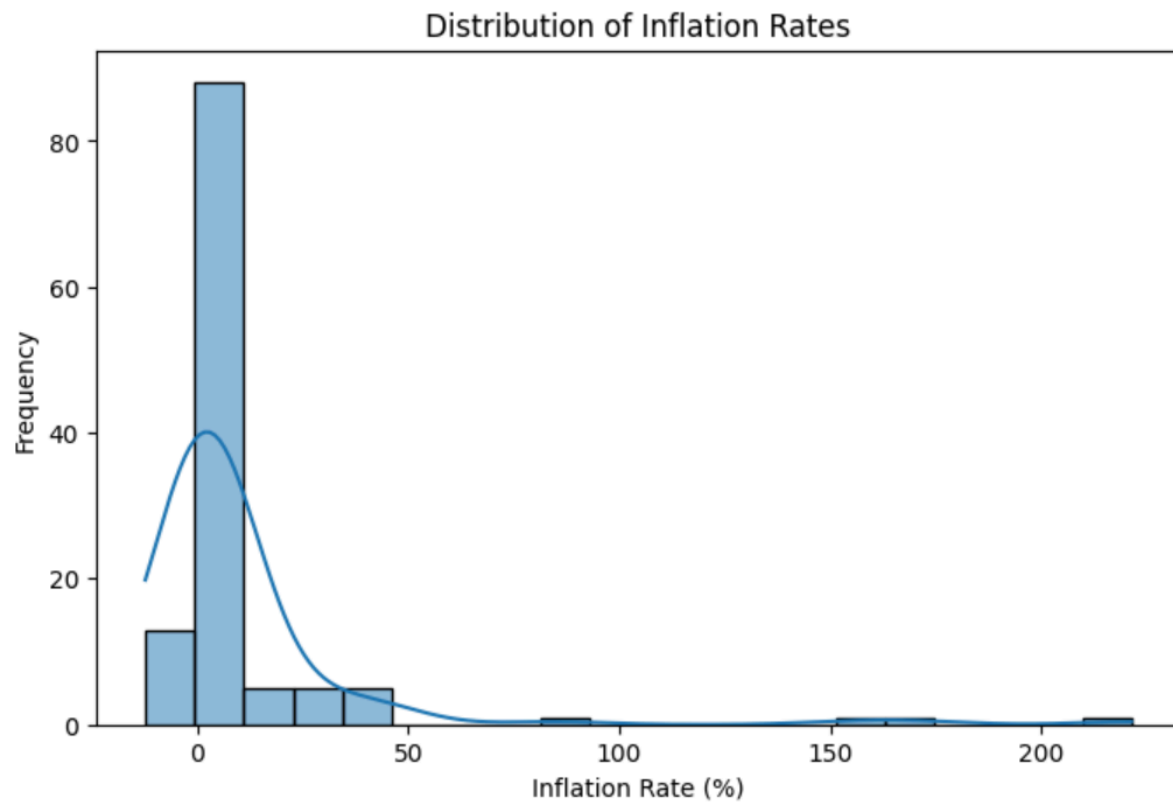
This scatter plot examines the relationship between Brent oil price and GDP growth. The plot suggests a weak positive relationship, where higher oil prices may be associated with higher GDP growth in some observations. This is reasonable because many Middle Eastern economies are directly or indirectly affected by oil revenues.

6.4 Distribution of Annual GDP Growth



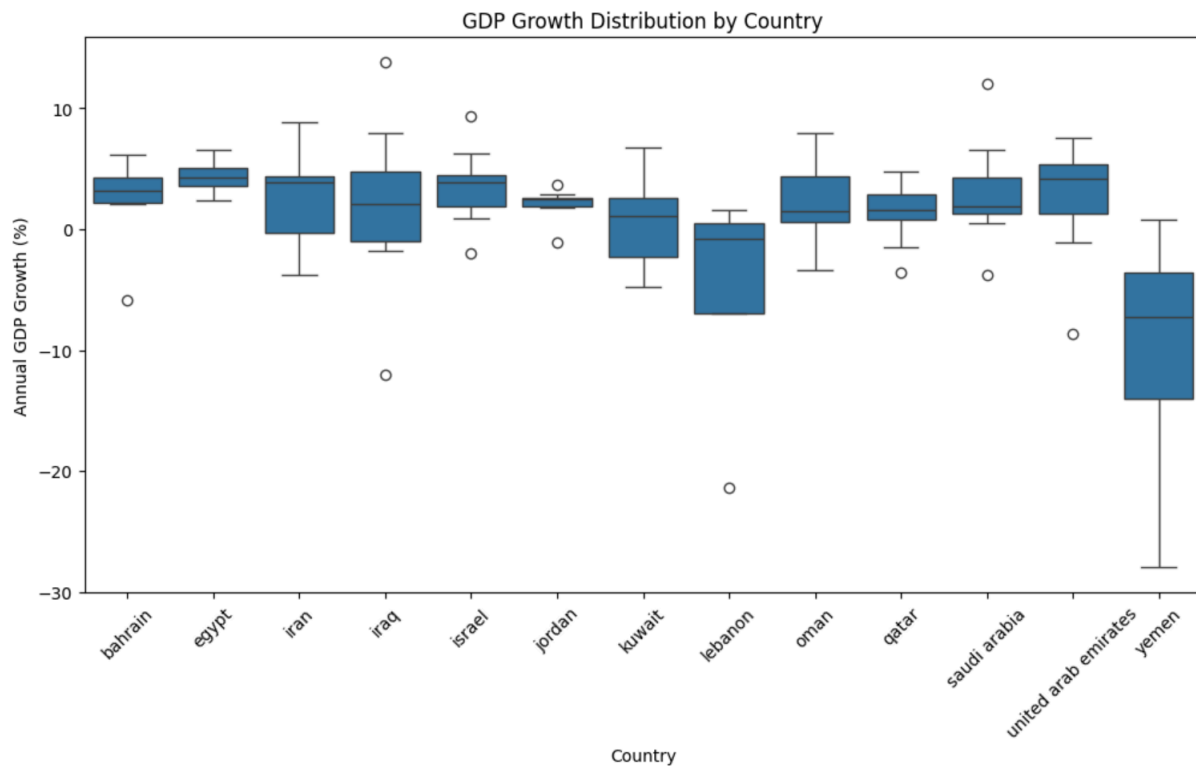
The histogram of GDP growth shows that most observations are concentrated around moderate GDP growth levels, but some observations show extreme negative or positive growth. These outliers reflect the volatility of economic performance in the region.

6.5 Distribution of Inflation Rates



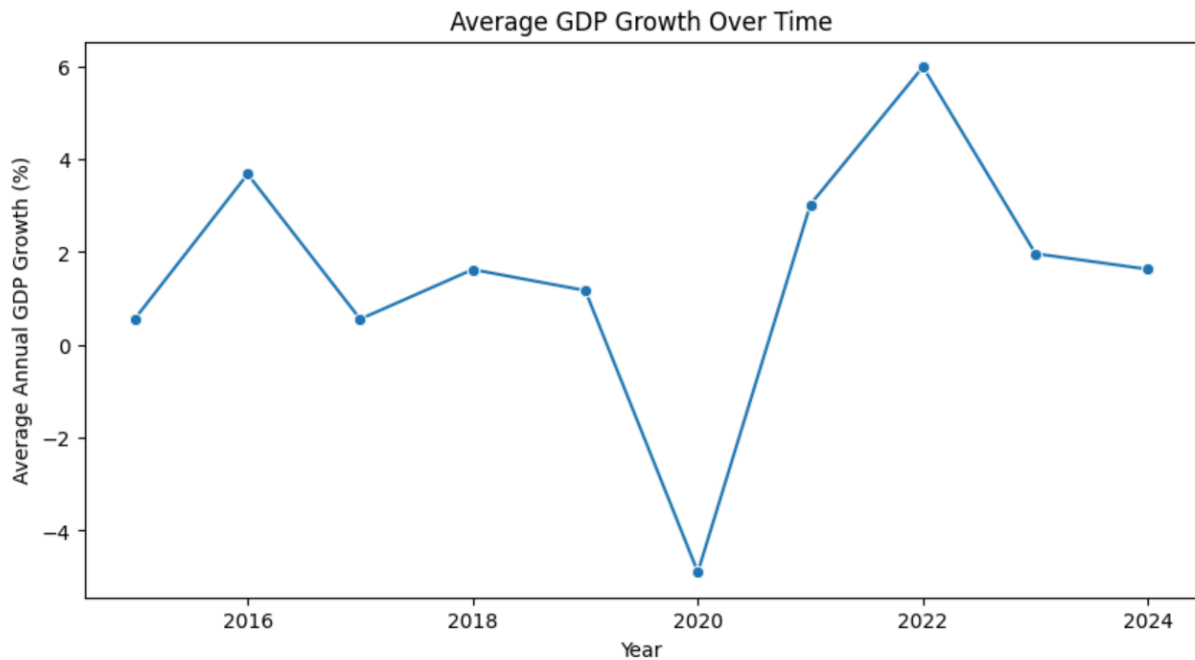
The inflation distribution is right-skewed. Most country-year observations have relatively low or moderate inflation, while a few observations show very high inflation. These high-inflation cases may represent periods of economic crisis, currency pressure, or political instability.

6.6 GDP Growth Distribution by Country



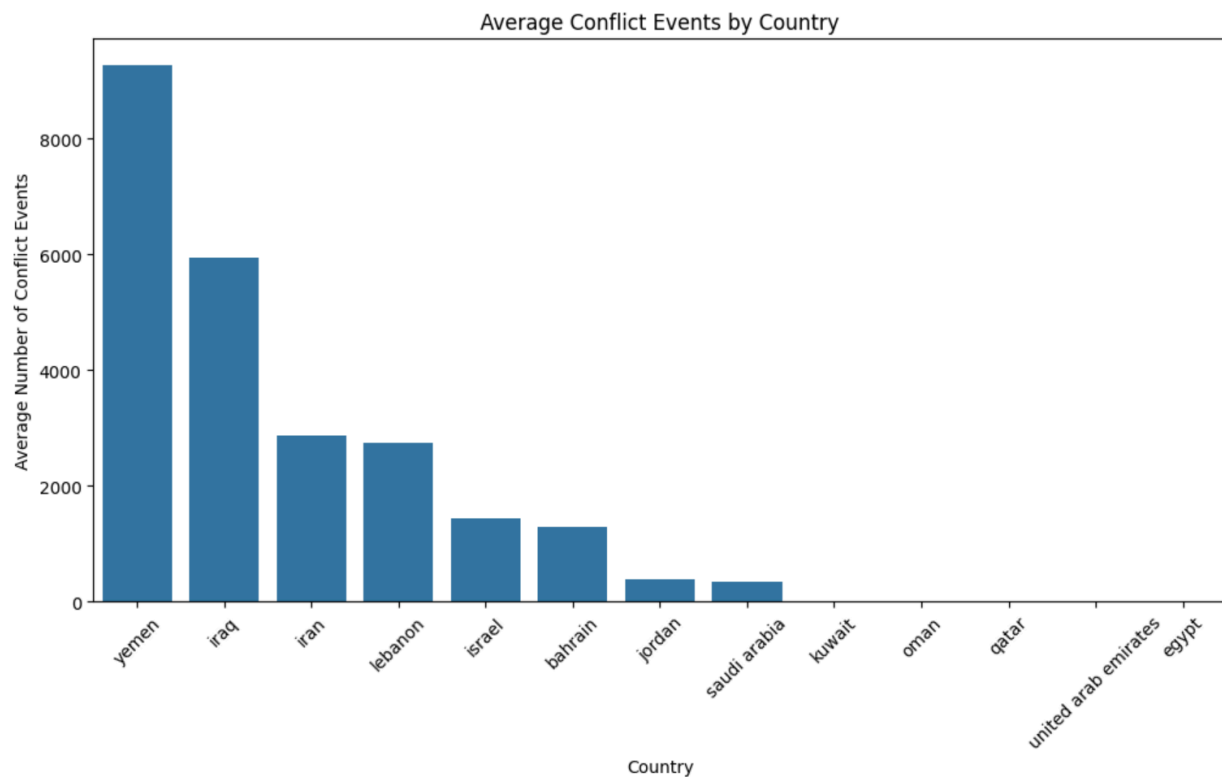
The boxplot compares GDP growth across countries. Some countries have relatively stable GDP growth, while others show wider variation and extreme outliers. This indicates that economic performance differs significantly across countries in the region.

6.7 Average GDP Growth Over Time



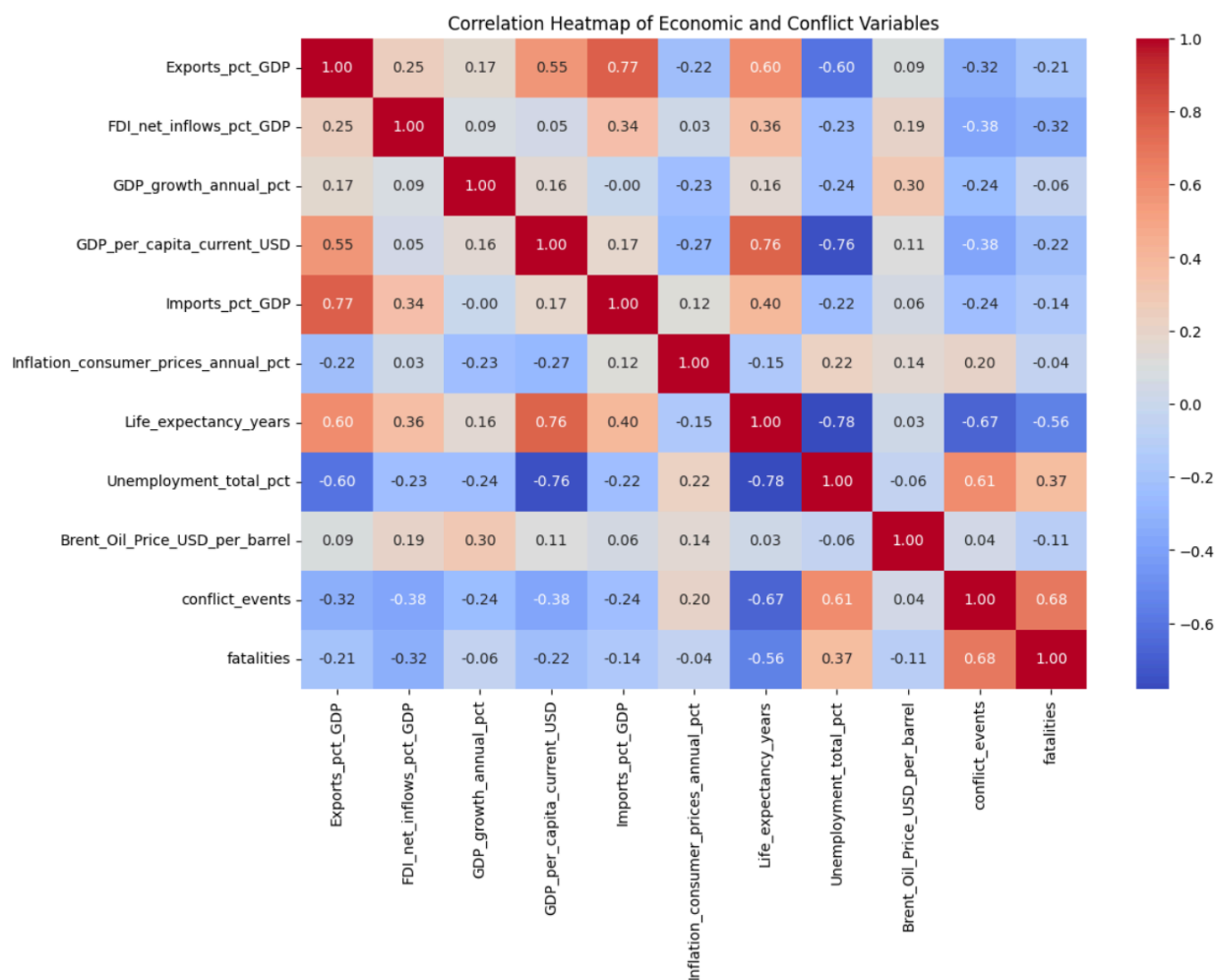
The line plot shows average GDP growth by year. The trend fluctuates over time, showing that regional growth changes significantly from year to year. Large drops may be connected to economic shocks, oil price changes, conflict, or global disruptions.

6.8 Average Conflict Events by Country



The bar chart compares average conflict events across countries. Countries with higher average conflict events contributed more heavily to the conflict variation in the dataset. This supports the idea that conflict intensity is not evenly distributed across the region.

6.9 Correlation Heatmap



The correlation heatmap summarizes relationships among economic indicators, oil price, conflict events, and fatalities. It helps identify which variables are positively or negatively related. This visualization is useful before hypothesis testing and machine learning because it provides an overview of potential relationships among variables.

7. Hypothesis Testing

Four hypotheses were tested using Pearson correlation analysis.

Hypothesis 1: Conflict Events and GDP Growth

- **H0:** Conflict has no effect on GDP growth.
- **H1:** Conflict negatively affects GDP growth.

The correlation between conflict events and GDP growth was approximately **-0.244**, with a p-value of approximately **0.0065**. This indicates a weak negative relationship that is statistically significant. Therefore, higher conflict events are associated with lower GDP growth.

Hypothesis 2: Fatalities and Inflation

- **H0:** Conflict fatalities do not affect inflation.
- **H1:** Conflict fatalities increase inflation.

The correlation between fatalities and inflation was approximately **-0.041**, with a p-value of approximately **0.654**. This result is not statistically significant. Therefore, the analysis does not provide evidence that fatalities are strongly related to inflation in this dataset.

Hypothesis 3: Conflict Events and FDI

- **H0:** Conflict has no effect on FDI.

- **H1:** Conflict reduces FDI.

The correlation between conflict events and FDI net inflows was approximately **-0.384**, with a very small p-value of approximately **0.0000118**. This suggests a statistically significant negative relationship. Countries and years with more conflict events tend to have lower FDI inflows as a percentage of GDP.

Hypothesis 4: Oil Price and GDP Growth

- **H0:** Oil price has no effect on GDP growth.
- **H1:** Oil price affects GDP growth.

The correlation between Brent oil price and GDP growth was approximately **0.299**, with a p-value of approximately **0.00077**. This indicates a weak to moderate positive relationship that is statistically significant. Higher oil prices are associated with higher GDP growth in this dataset.

8. Machine Learning Analysis

The machine learning part of the project focused on predicting annual GDP growth. Since GDP growth is a numerical variable, this is a supervised regression problem.

8.1 Features and Target Variable

The target variable was:

- Annual GDP Growth Percentage

The input features included:

- Exports as percentage of GDP
- FDI net inflows as percentage of GDP
- GDP per capita
- Imports as percentage of GDP
- Inflation
- Life expectancy
- Unemployment
- Brent oil price
- Conflict events
- Fatalities
- Log-transformed conflict events
- Log-transformed fatalities
- Year
- Country

Country was treated as a categorical variable and encoded using one-hot encoding. Missing numerical features were handled using median imputation.

8.2 Models Used

The following models were used:

1. **Baseline Mean Model**

This model predicts the mean GDP growth and is used as a comparison point.

2. **Linear Regression**

A simple regression model used to test whether a linear relationship can explain GDP growth.

3. **kNN Regression**

A non-parametric regression method that predicts by averaging nearby observations.

4. **Decision Tree Regression**

A tree-based model that learns rule-based splits from the data.

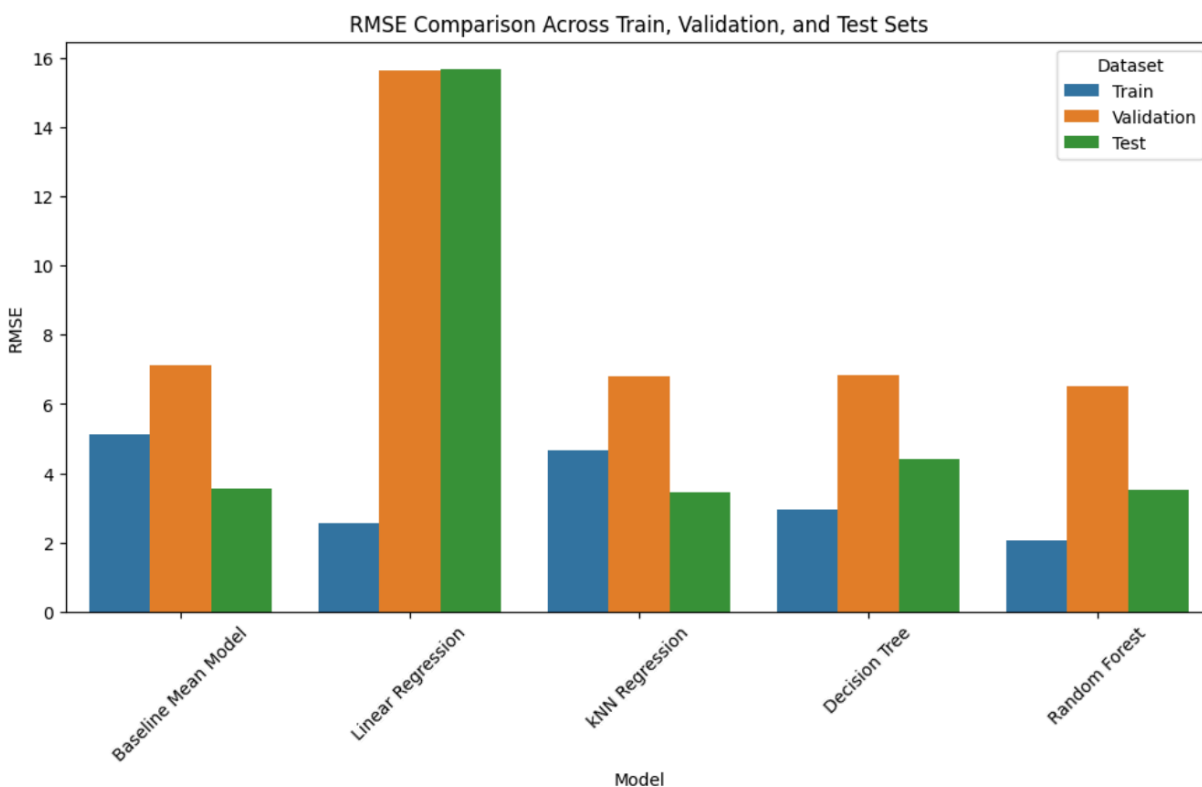
5. **Random Forest Regression**

An ensemble model that combines multiple decision trees.

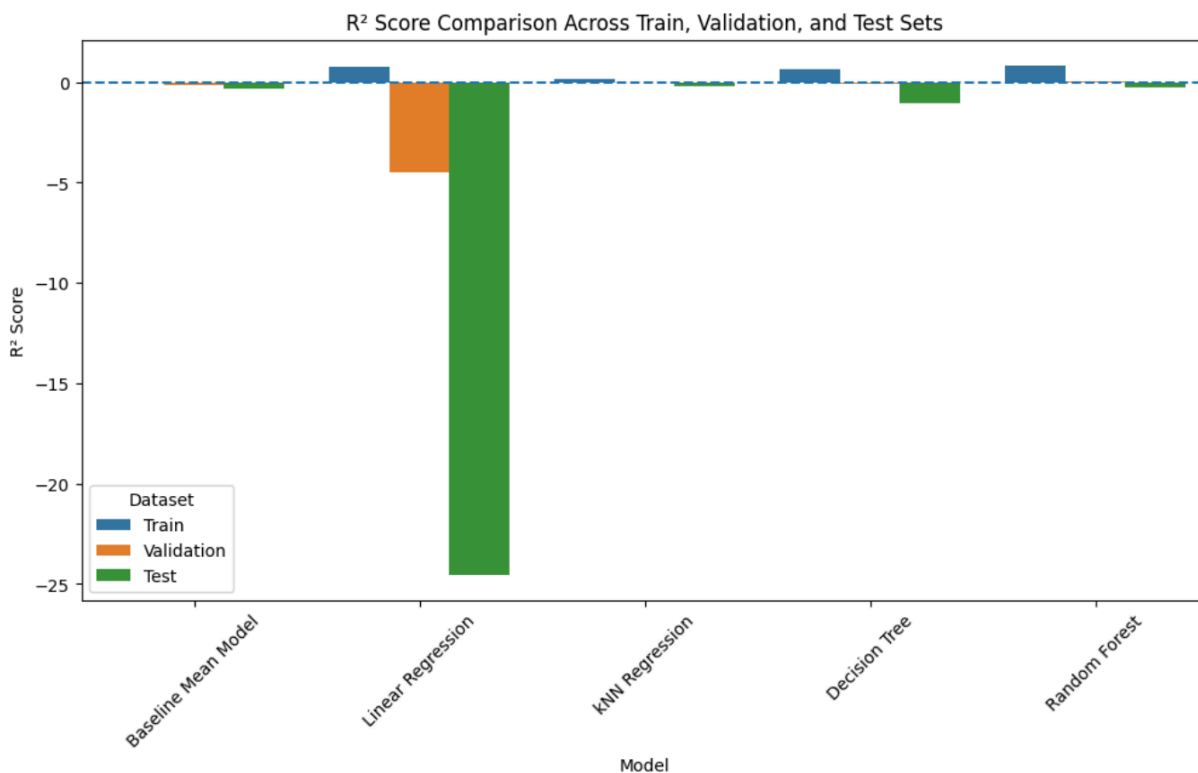
8.3 Hyperparameter Tuning

Hyperparameter tuning was applied to kNN Regression, Decision Tree Regression, and Random Forest Regression. A validation set was used to select better hyperparameters. This was done because hyperparameters such as the number of neighbors in kNN, tree depth, and number of estimators in Random Forest affect model complexity and performance.

8.4 Train / Validation / Test Evaluation



The models were compared across training, validation, and test sets. This was done to evaluate whether models generalized well to unseen data. A model that performs well on training data but poorly on validation or test data may be overfitting.



Several models had negative R² scores. A negative R² score means that the model performs worse than a simple baseline in explaining the variation in the target variable. This does not necessarily mean the implementation is wrong. Instead, it shows that GDP growth is difficult to predict using the available variables and limited dataset.

8.5 Best Model

The best-performing model on the test set was **kNN Regression**, based on RMSE. The final test set results were:

Model | MAE | RMSE | R^2 Score

kNN Regression | 2.5969 | 3.4323 | -0.2222

Random Forest | 2.6780 | 3.5068 | -0.2758

Baseline Mean Model | 2.9921 | 3.5459 | -0.3044

Decision Tree | 3.4050 | 4.4137 | -1.0211

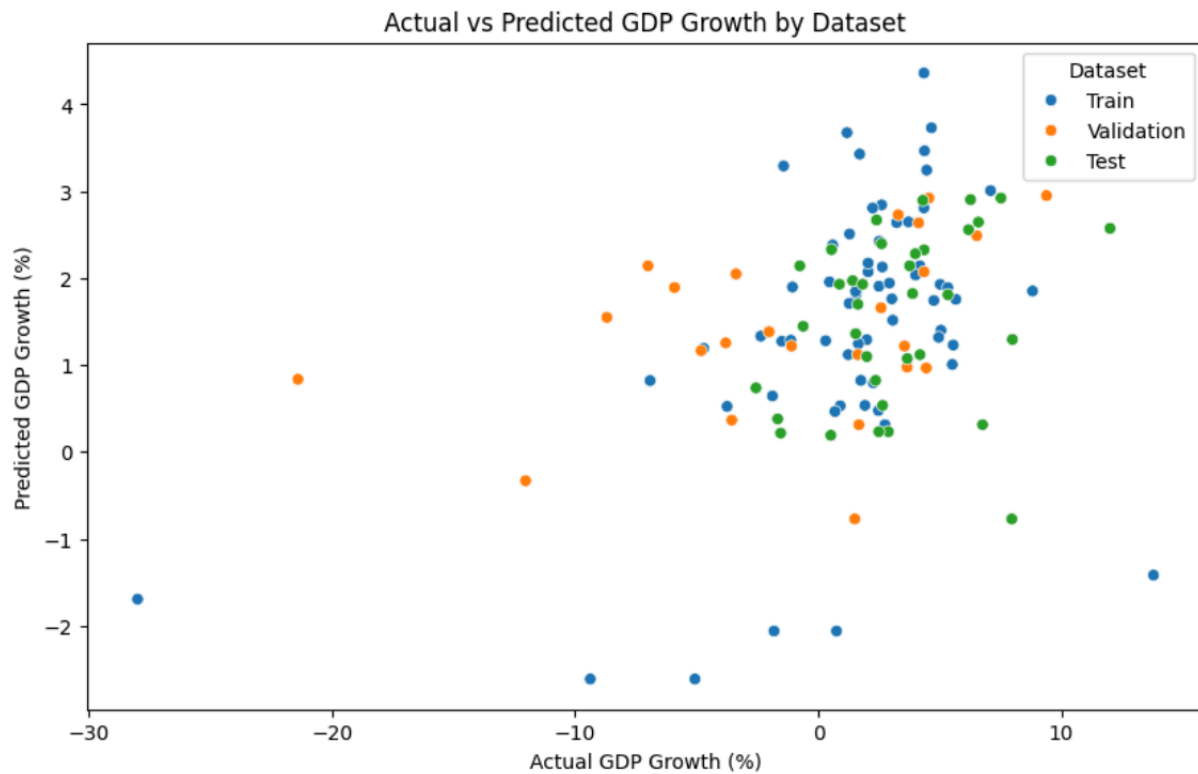
Linear Regression | 12.6230 | 15.6873 | -24.5312

Based on RMSE, kNN Regression performed best on the test set. However, all R^2 scores were negative, meaning the models had limited ability to explain unseen GDP growth values.

However, its R^2 score was still negative, indicating weak explanatory power on unseen data.

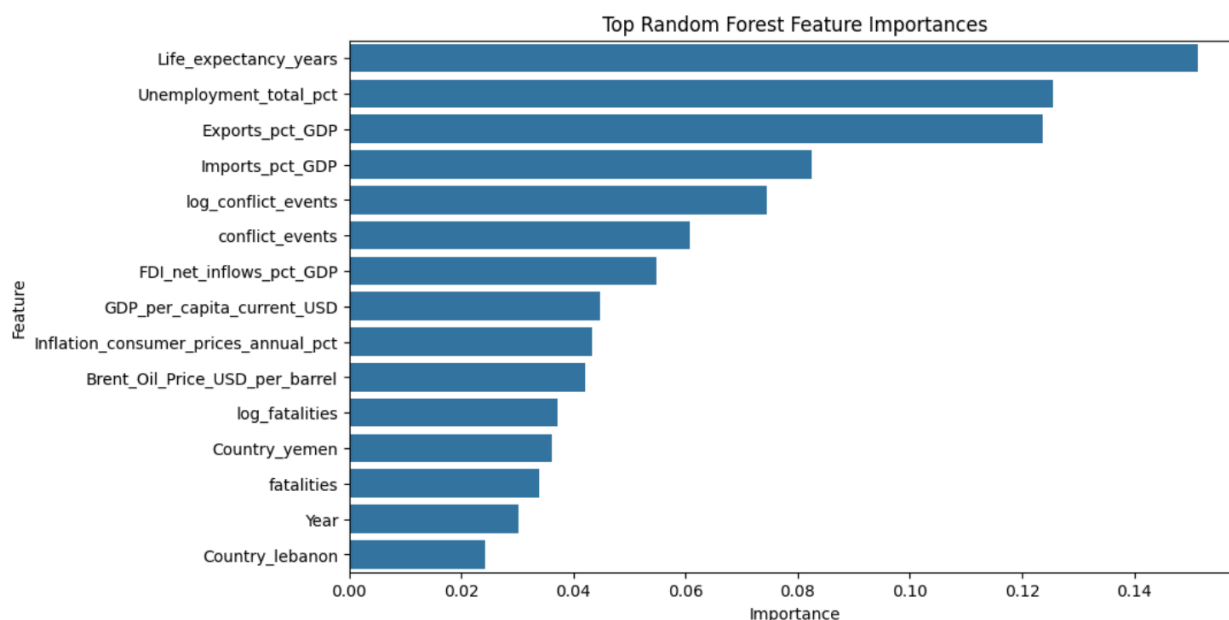
This means that although kNN Regression performed better than the other models in terms of RMSE, the overall predictive ability of the models remains limited.

8.6 Actual vs Predicted GDP Growth



The actual vs predicted plot compares model predictions across the training, validation, and test sets. A strong model would have points close to a diagonal pattern, where predicted values are close to actual values. In this project, the spread of points shows that the model does not fully capture GDP growth variation, especially for extreme values.

8.7 Random Forest Feature Importance



Random Forest feature importance was used to interpret which variables contributed most to GDP growth prediction. Since country was included as a categorical variable, one-hot encoded country features were also considered by the model. The feature importance results help identify whether macroeconomic variables, oil prices, conflict indicators, or country-level differences were more useful for prediction.

9. Findings

The project produced several important findings:

- Conflict events are negatively associated with GDP growth.
 - Conflict events are also negatively associated with FDI net inflows, suggesting that conflict may reduce investment attractiveness.
 - Fatalities do not show a statistically significant relationship with inflation.
 - Brent oil price is positively associated with GDP growth.
 - EDA shows that GDP growth and inflation have outliers, reflecting regional economic volatility.
 - Machine learning models had limited predictive power, shown by negative R2 scores.
 - kNN Regression performed best on the test set based on RMSE.
 - Random Forest feature importance suggests that GDP growth prediction depends on a mixture of macroeconomic variables, oil price, conflict variables, and country-level effects.
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10. Limitations and Future Work

This project has several limitations.

First, the final merged dataset is relatively small because the conflict dataset only overlaps with the economic data from 2015 to 2024. Machine learning models usually perform better with larger datasets, so the small number of country-year observations limits predictive accuracy.

Second, some economic variables have missing values. Although median imputation was used, missing data can still affect model performance and interpretation.

Third, correlation analysis cannot prove causality. The results show associations between variables, but they do not prove that conflict or oil prices directly cause changes in GDP growth.

Fourth, GDP growth is influenced by many factors not included in this project, such as sanctions, exchange rates, government spending, monetary policy, oil production, political reforms, and global crises.

Future work could improve the project by adding more variables, including oil production data, exchange rates, sanctions indicators, government spending, and political risk indices. More advanced time-series models could also be used to better handle the temporal nature of the data.

11. Ethical Considerations and AI Assistance

The datasets used in this project are publicly available and do not contain private personal information. However, conflict data represents sensitive real-world events involving political instability and human casualties. Therefore, the findings are interpreted carefully and not used to make unsupported causal claims.

AI assistance was used during the project for code organization, debugging, writing support, interpretation wording, and final report preparation. The code was run and reviewed by the student, and the final submission was controlled by the student.

A more detailed AI assistance disclosure, including the types of prompts used and the outputs generated, is included in the repository as `AI_ASSISTANCE.md`.

12. Conclusion

This project applied the full data science pipeline to study conflict, oil prices, and economic performance in the Middle East. The datasets were collected, cleaned, merged, explored visually, analyzed statistically, and used for machine learning.

The results suggest that conflict events are negatively associated with GDP growth and FDI, while oil prices are positively associated with GDP growth. However, machine learning results show that GDP growth is difficult to predict accurately using the available variables. The best-performing model was kNN Regression based on test RMSE, but negative R^2 scores indicate limited predictive power.

Overall, the project shows that both conflict and oil prices are relevant to economic outcomes in the Middle East, but GDP growth is complex and cannot be fully explained by the available data alone.